125101 - Analytical chemistry 1 for engineers

Tutorial 7– Acid/base Titration

T7-1. 100 mL of acetic acid (CH_3COOH) with a concentration of 0.05M (pKa = 4.76) was titrated with a 0.1M solution of NaOH. Calculate the pH at:

a. Before adding the base.

b. At 20% of the titration.

c. At 50% of the titration.

d. At 99% of the titration.

e. At the equivalent point

f. 0.5 mL excess of NaOH after the equivalent point.

g. 10 mL excess of NaOH after the equivalence point

h. Make a graph of pH as a function of the volume of the base that was added.

a) Before addition of the base,

$$CH_3COOH \leftrightarrow CH_3COO^- + H^+$$

$t = 0$	0.05	0	0
t = t	0.05 - x	x	x

$$Ka = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} = 10^{-4.76} = 1.74 \times 10^{-5}$$

$$Ka = \frac{x^2}{0.05 - x}$$

If f/Ka >100, the hydrolysis of $CH3COO^-$ could be neglected, which means $(0.05 - x) \approx 0.05$.

For this condition,

$0.05/1.74 \times 10^{-5} = 2.87 \times 10^3 >> 100$

So, it is safe to approximate $(0.05 - x)$ to 0.05.

$$x = \sqrt{Ka \cdot 0.05} = 9.33 \times 10^{-4} M$$
$$pH = -\log 9.33 \times 10^{-4} = 3.03$$

b) After 20% of the titration

$$pH = pKa + \log \frac{20}{80} = 4.16$$

c)After 50%

$$pH = pKa + \log \frac{50}{50} = 4.76$$

d)After 99%

$$pH = pKa + \log \frac{99}{1} = 6.76$$

e) At the equivalent point

There is just only CH_3COO^- in the solution, and the hydrolysis would produce OH^-.

$$CH_3COO^- + H_2O \leftrightarrow CH_3COOH + OH^-$$

V(NaOH) × 0.1M = 100 mL × 0.05 M CH_3COOH

So, V(NaOH) = 50 mL to get to the equivalent point.

At this position the total volume is 100 mL + 50 mL = 150 mL, so the concentrations are:

$$[CH_3COO^-] = \frac{100 \cdot 0.05}{150} = 0.03M$$

$$Kb = \frac{[CH_3COOH][OH^-]}{[CH_3COO^-]} = \frac{Kw}{Ka} = \frac{10^{-14}}{1.74 \times 10^{-5}} = 5.75 \times 10^{-10}$$

	$CH3COO^- + H_2O \rightleftharpoons$	$CH3COOH$	$+ OH^-$
t=0	*0.03*	*0*	*0*
t=t	*0.03-x*	*x*	*x*

If f/K_b >100, $0.03/5.75\times10^{-10} = 5.22 \times10^{7}$ >>100, 0.03-*x* is approximated to 0.03, and we can use the simple formula as below,

$$Kb = \frac{x^2}{0.03 - x}; \qquad x = \sqrt{0.03Kb} = 4.15 \times 10^{-6} M$$
$$pOH = -\log 4.15 \times 10^{-6} = 5.38$$
$$pH = 14 - 5.38 = 8.62$$

f) After the equivalent point, 0.5 ml of NaOH was added

$$[NaOH] = \frac{0.5 \cdot 0.1}{150 + 0.5} = 3.32 \times 10^{-4} M$$

$$pOH = -\log 3.32 \times 10^{-4} = 3.48$$
$$pH = 14 - 3.48 = 10.52$$

Let us check why the OH^- from the hydrolysis of CH_3COO^- can be neglected,

$$[OH^-] = c(NaOH) + \cancel{[A^-]}$$

	$CH3COO^-$ + H_2O ⇌	CH3COOH	+ OH^-
t=0	*0.03*	*x*	*x*
t=t	*0.03-x*	*x*	*x+3.32×10*$^{-4}$

$$\frac{(x+3.32\times10^{-4})x}{0.03-x} = 5.75\times10^{-10}$$

$$x^2 + 3.32\times10^{-4} - 1.72\times10^{-11} = 0$$

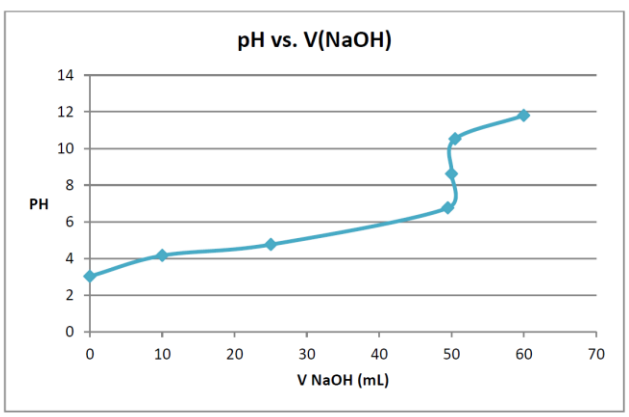
The value of $x = 5.18\times10^{-8}$, which is too small compared to 3.32×10^{-4}, which means the OH^- from the hydrolysis of CH_3COO^- in the solution can be omitted.

g) After equivalent point, 10 ml NaOH was in excess

$$[NaOH] = \frac{10 \cdot 0.1}{150 + 10} = 6.25 \times 10^{-3} M$$

$$pOH = -\log 6.25 \times 10^{-3} = 2.20$$
$$pH = 14 - 2.20 = 11.80$$

h) Graph of pH as a function of the volume of NaOH,



T7-2. 50 mL of the base methylamine (CH_3NH_2) with a concentration of 0.042M was titrated with HCl and this are the results:

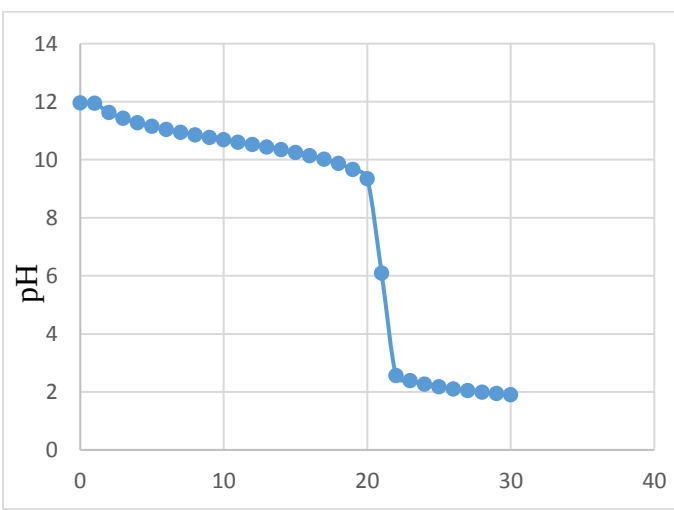
V(HCl)	pH	V(HCl)	pH	V(HCl)	pH
0	11.959	10	10.686	20	9.344
1	11.946	11	10.604	21	6.087
2	11.623	12	10.52	22	2.556
3	11.423	13	10.434	23	2.386
4	11.273	14	10.344	24	2.267
5	11.15	15	10.247	25	2.176
6	11.043	16	10.14	26	2.103
7	10.946	17	10.017	27	2.041
8	10.856	18	9.867	28	1.989
9	10.77	19	9.667	29	1.943

Given that the reaction is: $CH_3NH_2 + HCl \longrightarrow CH_3NH_3^+ + Cl^-$

a. What is the K_a of $CH_3NH_3^+$ (What is the K_b for the methylamine CH_3NH_2)?

b. What is the concentration of the HCl?

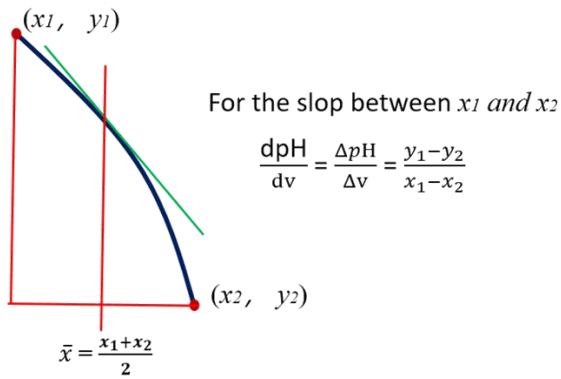
The graph of pH as a function of the volume of HCl can be obtained by Excel,



V HCl (mL)

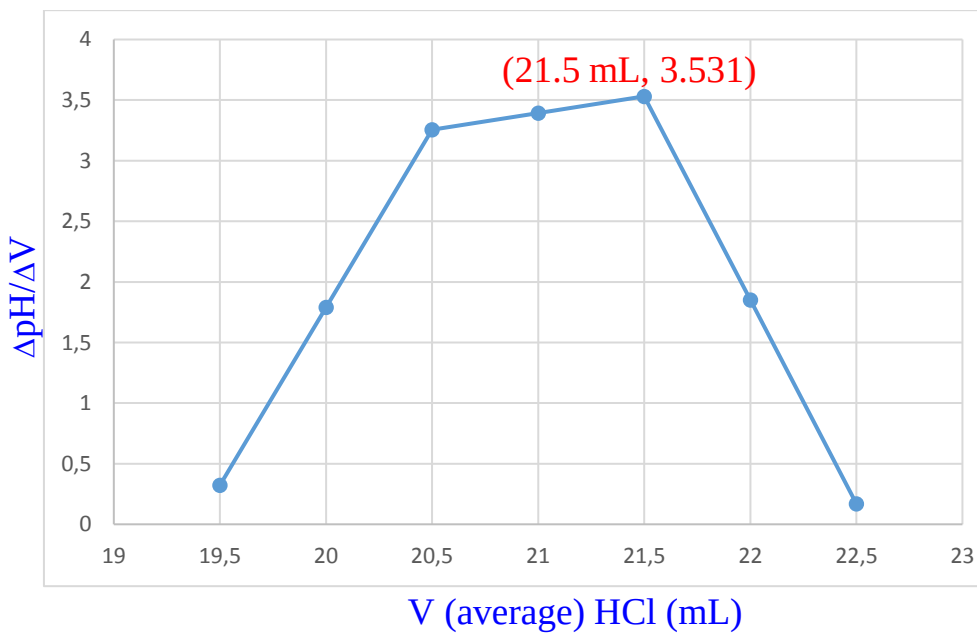
The sudden change of the pH locates at the range of 20 ~ 22 mL.

V(HCl)	pH
19	9.667
20	9.344
21	6.087
22	2.556
23	2.386

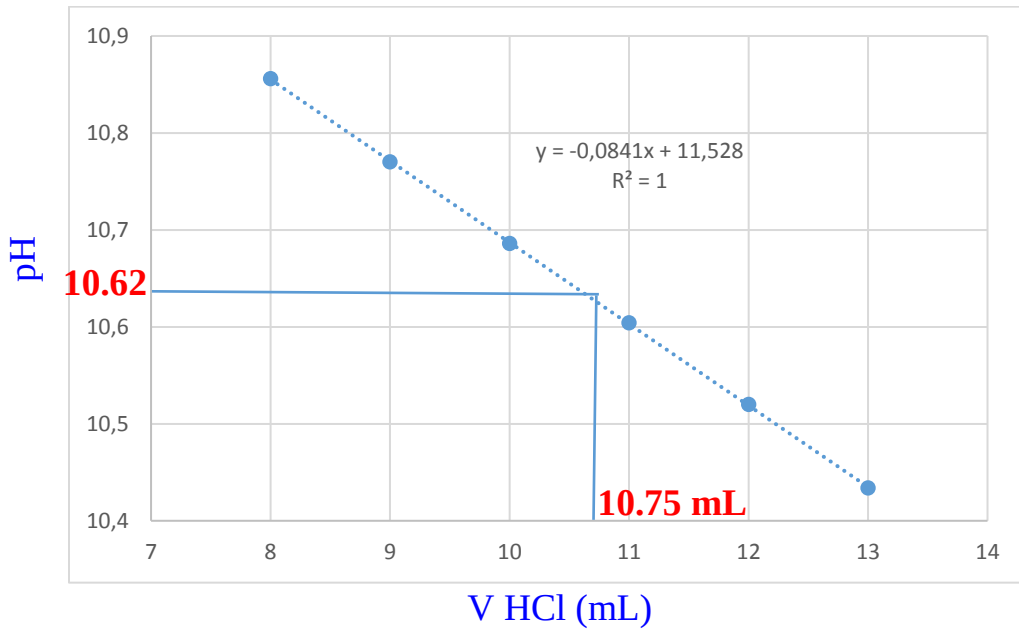


We can obtain the different slopes with the difference of HCl volume,

$\bar{v}$	Slope ($\Delta pH/\Delta V$)
19.5	0.323
20.0	1.79
20.5	3.257
21.0	3.394
21.5	3.531
22.0	1.850
22.5	0.170



a). From the diagram, V_e = 21.5 mL can be obtained, **the volume under the maximum of $\frac{dpH}{dV}$ (the first derivative of the titration curve),** which means the equivalence point for the volume of HCl. The pK_a is the pH at half volume of the titrant. That is to say, when $V_a = \frac{1}{2}V_e$ = 21.50 mL/2 = 10.75 mL. The pH after adding 10 mL is 10.686, after adding 11 mL the pH is 10.604. We can use the external standard (interpolation) to get the unknown pH at V_a = 10.75 mL.



So, the $CH_3NH_3^+$ pK_a is the pH at the HCl volume of 10.75 mL, which is 10.62.

You also can use this formula to get pK_a

pKa = 10.686 + 0.75* (10.604-10.686) = 10.6245

pK$_b$ for methylamine CH_3NH_2 is pK$_b$ = 14 – pKa = 3.38

b) What is the concentration of the HCl?

$$CH_3NH_2 + HCl \rightarrow CH_3NH_3^+ + Cl^-$$

At the equivalent point, the mole of HCl was exactly equal to the mole of the base because the ration for the reaction is 1:1. So, when V_e = 21.5 mL,

$$n(CH_3NH_2) = 0.042M \cdot 50mL = 2.1mmol = n(HCl)$$

$$[HCl] = \frac{2.1mmol}{21.5mL} = 0.098M$$

T7-3. Calculate the titration error if we stop the titration of V_0 = 50.0 mL of a C_0 = 0.10 M solution of HA with $K_a = 5\times10^{-5}$ at end point pH = 7.52 instead of the theoretical pH= 8.50. The titration is performed with a C = 0.10 M solution of NaOH.

Applying the equation

$$\text{Titration error} = \frac{C+C_0}{CC_0}\left[\frac{Kw}{[H^+]_{ep}} - [H^+]_{ep}\right] - \frac{[H^+]_{ep}}{Ka}$$

For C and C_0 is the same 0.10 M.

At end point, $[H^+]_{ep} = 10^{-7.52}$ M

$Ka = 5\times10^{-5}$

We obtain (at end point pH = 7.52)

$$\text{Titration error} = \varphi_{en} - 1 = \frac{0.10+0.10}{0.10\times0.10}\left(\frac{10^{-14}}{10^{-7.52}} - 10^{-7.52}\right) - \frac{10^{-7.52}}{5\times10^{-5}}$$

$$= -5.96\times10^{-4}$$

The relative percent error is 0.06%.

This error is smaller than the error associated with measuring the volume (reading error of burette is about 0.1%), which means the selected indicator is suitable for this titration.

Think about what's the meaning of the titration errorsign (-). It means the error have direction.

The error at equivalence point (theoretical pH= 8.50) also can be calculated,

$$\text{Titration error} = \varphi_{en} - 1 = \frac{0.10+0.10}{0.10\times0.10}\left(\frac{10^{-14}}{10^{-8.50}} - 10^{-8.50}\right) - \frac{10^{-8.50}}{5\times10^{-5}}$$

$$= 20 \times (3.15884\times10^{-6}) - 6.3246\times10^{-5}$$

$$= 6.3177\times10^{-5} - 6.3246\times10^{-5}$$

$$= -7\times10^{-8} \approx 0$$

Which mean $\varphi_{ep} - 1 = 0 \Longrightarrow \varphi_{ep} = 1$

Then, $\varphi_{ep} = \frac{Cv_{ep}}{C_0v_0} = 1$

So, $CV_{ep} = C_0V_0$